

MUHAMMAD SHAHAL

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PROFESSIONAL SUMMARY

Full-stack AI/ML Engineer with production experience building and deploying scalable AI applications and software systems. Experienced in Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI. Skilled in **backend development** (FastAPI, PostgreSQL, DynamoDB, AWS S3, vector databases), **frontend development** (React), and **cloud-native deployment** (AWS, Docker). Proficient in Python, LangChain, and LangGraph, with a focus on reliable, scalable, end-to-end systems.

PROFESSIONAL EXPERIENCE

Sustainability Economics.ai — Generative AI Engineer

2026

- Built an **LLM-driven** Grid Tariff Data Automation platform extracting and standardizing electricity tariff data from utility PDFs and websites across countries (**AWS S3, DynamoDB**), with automated **LLM extraction, evaluation, and Pydantic** schema validation — including country-specific mapping and unit conversion — cutting manual effort by **~80%**.
- Fine-tuned** a **Gemma 3 4B** model using **Unsloth** on domain-specific electricity tariff data to improve structured tariff extraction and enable efficient, cost-effective inference for product development.

Cybpres Innovative Solutions — AI Developer & Data Scientist

2025-2026

- Built and deployed **agentic AI systems** using **LangChain** and **LangGraph** enabling multi-step reasoning, tool calling, persistent conversational memory, and production-grade deployment.
- Engineered **Retrieval-Augmented Generation (RAG)** and hybrid search pipelines on **OpenSearch/FAISS** (BM25 + vector retrieval with reranking), containerized with **Docker** and optimized for high-performance inference.

Luminar Technolab — Data Science Intern

2024-2025

- Built end-to-end **machine learning pipelines** with automated **CI/CD** via **GitHub Actions**, supporting model training, evaluation, and deployment workflows.
- Developed **Natural Language Processing (NLP)** semantic-similarity models for classification and retrieval (~25% accuracy gain), and delivered **Power BI/Tableau** dashboards that cut manual reporting effort by ~50%.

KEY PROJECTS

Multi-Agent Legal Reasoning System (LLM + RAG + Persistent Memory)

- Architected a constraint-driven, **BNS-aligned** legal reasoning system using a global immutable retrieval index to prevent IPC/BNS statutory mixing and reduce hallucination.
- Orchestrated a **multi-agent LangGraph pipeline** with Postgres-backed persistent memory, checkpointed execution, and async **FastAPI** backend featuring token streaming and thread-isolated **ChromaDB** vector stores.

AI RAG MedBot — Healthcare Knowledge Assistant

- Built a **RAG medical chatbot** (LangChain, Groq LLaMA-3, Pinecone, Hugging Face embeddings) for precise Q&A over hospital CSVs and PDF texts.
- Deployed a scalable **Flask + Streamlit** pipeline, improving response relevance by ~30%, and made it reusable by any hospital via a simple CSV/dataframe swap.

Resume Matcher Pro — Full-Stack Parsing & Assessment System

- Developed a full-stack **resume parser** (FastAPI, React, PostgreSQL) to extract key candidate data efficiently.
- Implemented **NLP-based scoring algorithms** for real-time resume-job alignment assessment, improving matching accuracy and surfacing instant insights.

TECHNICAL SKILLS

Languages, Backend & Frontend: Python, SQL (PostgreSQL / MySQL), React, Bash, FastAPI, Flask, RESTful APIs, System Design

Generative AI & LLMs: LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, OpenAI SDK, LLaMA, Vector Databases (Pinecone, FAISS, ChromaDB, OpenSearch), Embeddings, Semantic & Hybrid Search

Machine Learning & NLP: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, Natural Language Processing (NLP), PEFT, LoRA, QLoRA, TensorRT, ONNX Runtime, Model Evaluation

Cloud & MLOps: AWS, Azure, Docker, CI/CD (GitHub Actions), MLflow, DVC, Inference Optimization, Airflow

Data Engineering & Visualization: ETL, Pandas, NumPy, RabbitMQ, Redis, Power BI, Tableau, Matplotlib, Seaborn

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning

Hindusthan College of Engineering and Technology | 2020-2024

Certifications: IBM Industry Certification Program | Data Science (NACTET) | Learnathon ICT